

2.4 FORMATTING ANALOG INFORMATION

| The Sampling Theorem

- Sampling Theorem is a theory that defines the sampling rate required when converting an analog signal (a continuous signal) into a digital signal (a computer-readable signal), specifying how frequently samples must be taken.
- This theory is also widely known as the **Nyquist-Shannon Sampling Theorem**.

Formula:

$$f_s \geq 2 f_{\max}$$

Where: f_s = Sampling Frequency
 $f_{\max}$ = Maximum Signal Frequency

| The Sampling Process

Example:

Maximum signal frequency = 20 kHz

Minimum sampling frequency: $2 \times 20 = 40$ kHz

In practice, Audio CDs use 44.1 kHz.

| Types of Sampling

- ✓ Under sampling
 - Sampling frequency is below the Nyquist rate
 - Causes aliasing
- ✓ Nyquist Sampling
 - Sampling frequency equals twice the highest signal frequency.
 - Allows accurate reconstruction.
- ✓ Over Sampling
 - Sampling frequency is greater than the Nyquist rate.
 - Improves signal quality and simplifies filtering.

Overview Comparison Table

Condition	Sampling Rate (f_s)	Result / Outcome	Practical Impact
Undersampling	$f_s < 2f_{\max}$	Aliasing occurs	Data is distorted; original signal cannot be reconstructed.
Nyquist Sampling	$f_s = 2f_{\max}$	Perfect Reconstruction	Theoretically perfect, but difficult to implement in practice due to ideal filter requirements.
Oversampling	$f_s > 2f_{\max}$	Improved Quality & Simpler Filtering	The most common and reliable approach used in practical digital systems.

- Nyquist Rate (f_N) = $2 \times f_m$ (where f_m is the maximum frequency of the signal)
- Nyquist Interval (T_N) = $\frac{1}{f_N} = \frac{1}{2f_m}$

Problem 1: Basic Single-Frequency Signal

Question: Find the **Nyquist Rate and Nyquist Interval** for the signal $x(t) = 5 \cos(2000\pi t)$.

Solution:

1. Compare the given signal with the standard formula $x(t) = A \cos(2\pi f_m t)$:

$$2\pi f_m t = 2000\pi t$$

$$2f_m = 2000$$

$$f_m = 1000 \text{ Hz (or 1 kHz)}$$

2. Calculate the **Nyquist Rate** (f_N):

$$f_N = 2 \times f_m = 2 \times 1000 = \mathbf{2000 \text{ Hz (or 2 kHz)}}$$

3. Calculate the **Nyquist Interval** (T_N):

$$T_N = \frac{1}{f_N} = \frac{1}{2000} = \mathbf{0.0005 \text{ seconds (or 0.5 ms)}}$$

Problem 2: Composite Signal (Multiple Frequencies Added)

Question: Find the Nyquist Rate for the signal $x(t) = 3 \cos(500\pi t) + 4 \sin(1000\pi t) + 10 \cos(3000\pi t)$.

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Solution:

- Find the frequency of each individual component:
 - First component: $2\pi f_1 = 500\pi \Rightarrow f_1 = 250 \text{ Hz}$
 - Second component: $2\pi f_2 = 1000\pi \Rightarrow f_2 = 500 \text{ Hz}$
 - Third component: $2\pi f_3 = 3000\pi \Rightarrow f_3 = 1500 \text{ Hz}$
- According to the Sampling Theorem, choose the **maximum frequency** (f_m):

$$f_m = \max(250, 500, 1500) = 1500 \text{ Hz}$$

- Calculate the **Nyquist Rate** (f_N):

$$f_N = 2 \times f_m = 2 \times 1500 = \mathbf{3000 \text{ Hz (or 3 kHz)}}$$

Problem 3: Multiplication of Two Signals

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Solution:

Note: When two signals are multiplied in the time domain, their frequencies add up in the frequency domain ($f_m = f_1 + f_2$).

1. Find the frequency of each signal:

- $2\pi f_1 = 200\pi \Rightarrow f_1 = 100 \text{ Hz}$

- $2\pi f_2 = 400\pi \Rightarrow f_2 = 200 \text{ Hz}$

2. Find the maximum frequency (f_m) of the multiplied product:

$$f_m = f_1 + f_2 = 100 + 200 = 300 \text{ Hz}$$

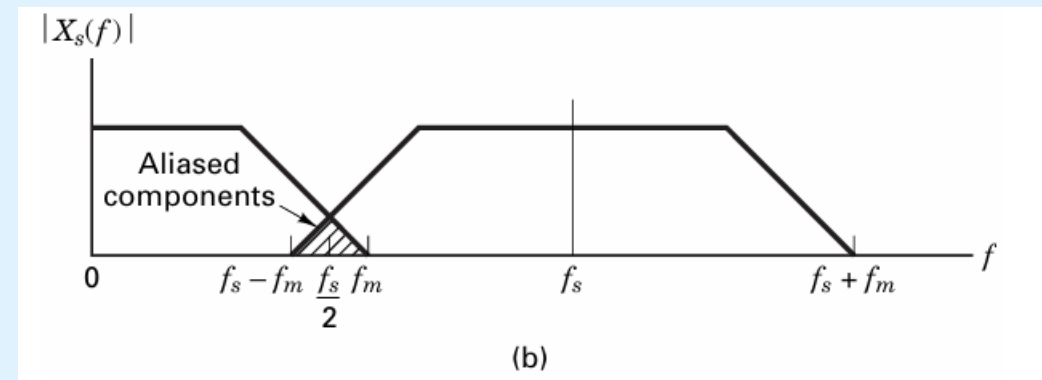
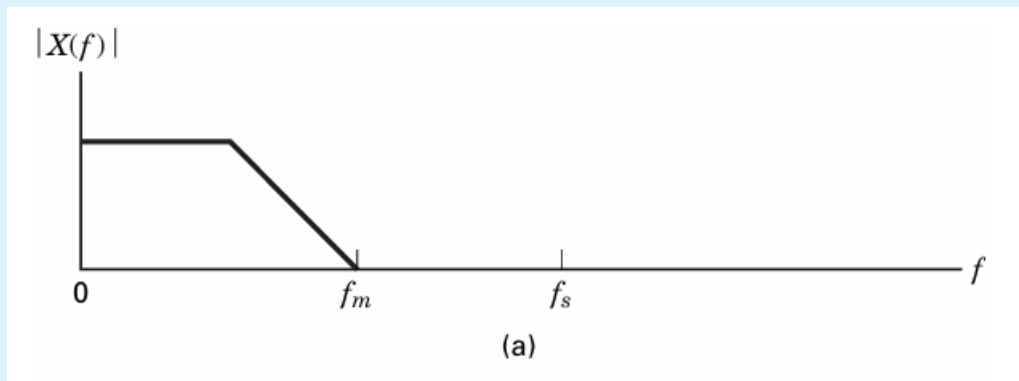
3. Calculate the **Nyquist Rate** (f_N):

$$f_N = 2 \times f_m = 2 \times 300 = \mathbf{600 \text{ Hz}}$$

Question: Find the Nyquist Rate for $x(t) = 5 \sin(600\pi t) + 2\cos(800\pi t)$.

Aliasing

- ✓ The "Alias" Concept: When undersampled, high-frequency signals masquerade as lower-frequency "aliases".
- ✓ Spectral Overlapping: In the frequency domain, the spectral replicates (copies of the spectrum) overlap with each other.
- ✓ Ambiguous Data: The overlapped region contains distorted data that appears in the frequency band between $(f_s - f_m)$ and f_m .



| Quantization

- Quantization is the process of converting the continuous amplitude (the high/low values) of an analog signal into discrete, predefined levels that can be utilized by a digital system.

There are **3 main steps** in an **ADC (Analog to Digital Converter)**:

Sampling – Taking samples of the continuous signal at specific intervals of time.

Quantization – Mapping or rounding the sampled amplitudes to the nearest predefined discrete level.

Encoding – Converting those quantized levels into digital binary code (0s and 1s).

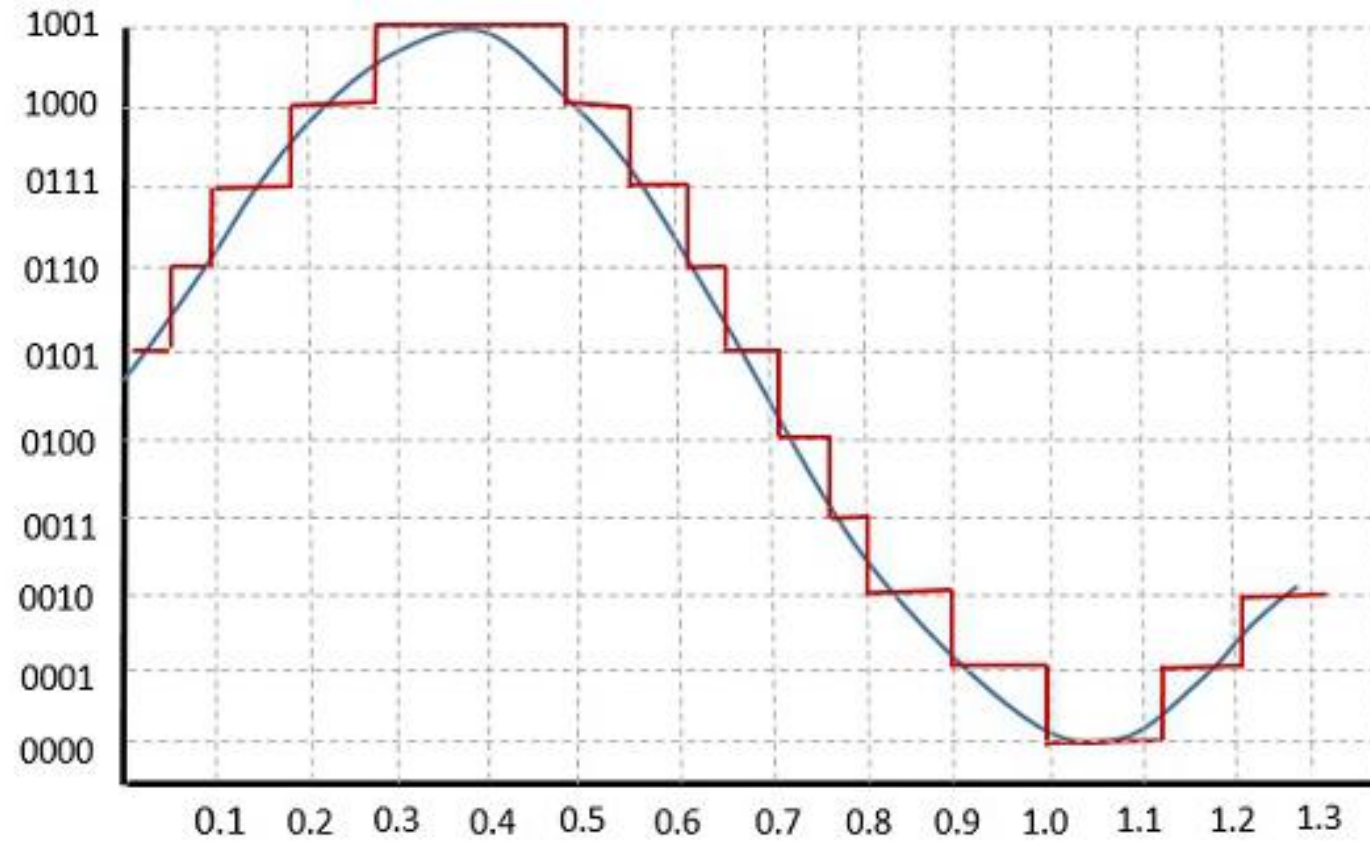
| Quantizing an Analog Signal

Analog-to-Digital Converter (ADC) converts analog data (such as waves or sound) into digital values (the 0s and 1s that computers understand), it goes through two main steps: **Sampling** and **Quantizing**.

Sampling: An analog wave flows continuously without interruption. Sampling is the process of capturing and recording specific data points (sample values) along that continuous wave at predetermined time intervals.

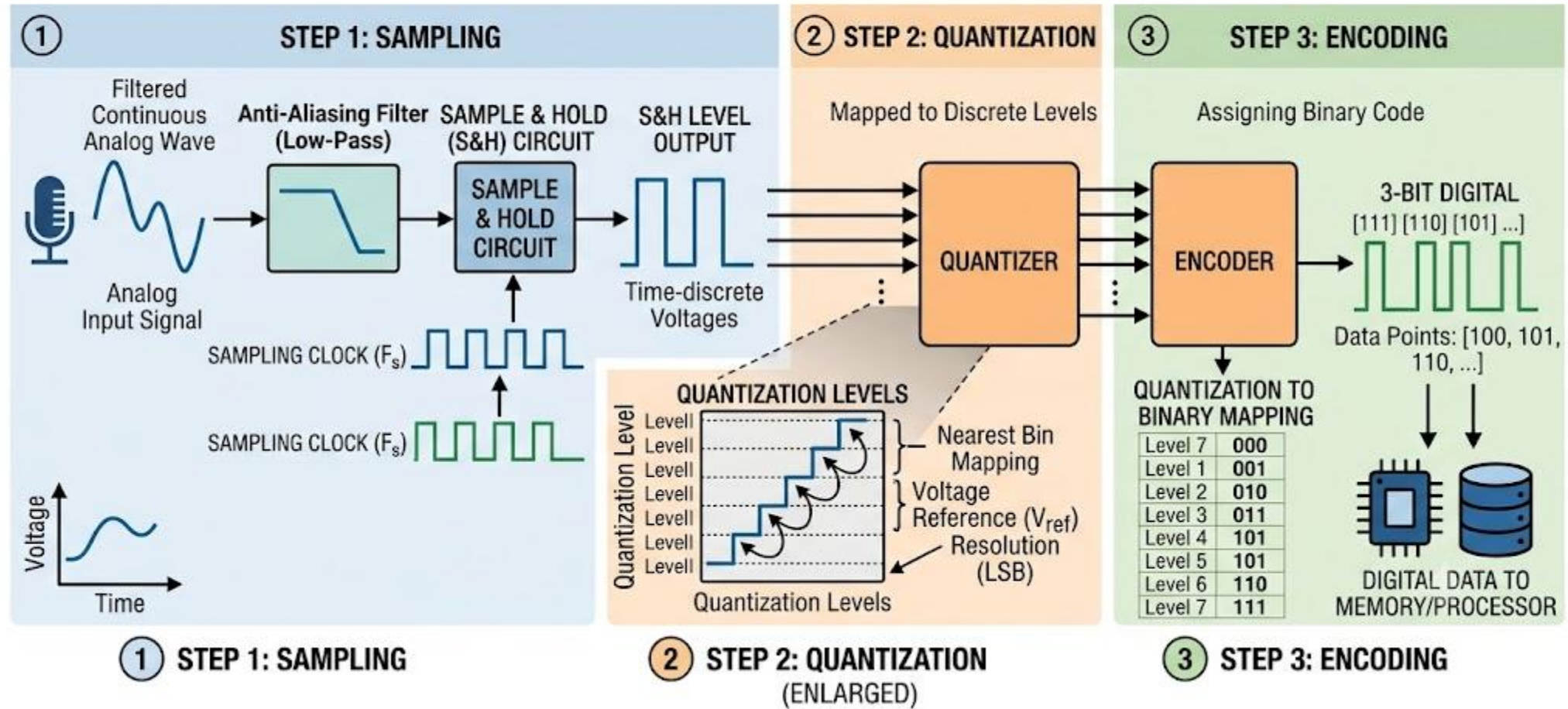
Quantizing: Quantizing is the step where the continuous voltage values of the dots obtained from sampling are assigned specific, exact digital levels.

Analog-to-Digital Conversion (ADC) process graph



3-BIT ADC PROCESS FLOW (3 FUNDAMENTAL STEPS)

ANALOG INPUT: Filtered continuous signal (e.g., from a microphone)



| PCM (Pulse Code Modulation)

- ✓ PCM is the standard and most widely used technique for Analog-to-Digital Conversion (ADC).
- ✓ It converts a continuous analog signal (such as a human voice or music) into a stream of digital binary bits (0s and 1s) without changing the core information.

The PCM process consists of three distinct steps:

Sampling: The continuous analog signal is measured at uniform time intervals. According to the Nyquist-Shannon Theorem, the sampling rate (f_s) must be at least twice the highest frequency component ($f_{\max}$) of the original signal ($f_s \geq 2f_{\max}$) to avoid distortion.

Quantization: The continuous amplitude values obtained from sampling are rounded off to the nearest pre-defined discrete voltage levels (Quantization Levels). This step introduces a minor deviation known as quantization noise.

Encoding: Each quantized level is mapped into a specific binary code (e.g., a 3-bit system creates codes from 000 to 111).

| Baseband Transmission

Once PCM produces the digital bitstream, it must be transmitted across a communication channel. There are two primary transmission methods: Baseband and Broadband (Passband).

Baseband Transmission is defined by the following characteristics:

The digital signal is transmitted over the physical medium (cables) directly without modulation (i.e., without shifting the signal to a higher frequency band).

The signal retains its original frequency range, and it utilizes the entire bandwidth of the channel exclusively for a single transmission session.